

Research papers

Thermal performance improvement of a heat-sink using metal foams for better energy storage systems

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ABSTRACT

Newly, energy storage systems and harvesting energy equipment are modern and important topics in engineering problems, but they still face the efficiency problem. Using the advantages of micro-scale flow and metal porous materials can be applied to improve the performance of these systems. So, the present investigation provides a numerical study on a heatsink employing various arrangements of metal foams inside a heat sink (in the direction and against the fluid flow). The influences of diverse fluid flow characteristics, metal foams' permeability and thermophysical properties of the coolant (using CuO-MgO/water nanofluids) on the thermal efficiency of the heatsink are examined. Then, the variations of convective heat transfer coefficient and pressure loss for all variables are investigated, and performance evaluation criteria are defined to distinguish the simultaneous impacts of improving heat exchange against incrementing the pressure loss. It indicates the metal foams located in the direction of the flow perform better than those located against the flow, which increases the performance evaluation criteria (PEC) of the heatsink between 1.18 and 1.55 depending upon the permeability of porous media. Also, using the nanofluids in the presence and absence of metal foams decreases the heatsink performance. Investigating the impacts of volumetric flow rate illustrates that thermal performance increases by increasing the flow rate. In addition, outcomes reveal that although adding metal foams and nano-additives and increasing volumetric flow rate enhance the frictional entropy generation rate, the total amount of entropy generation rate decreases due to more contribution of the thermal entropy generation rate, that reduces by those variables.

1. Introduction

A wide range of attention has recently been paid to energy storage systems due to various applications in biomedical, electronic cooling, ventilation and solar collectors industries [1–4]. One of the biggest challenges in the electronic industry can be addressed by the cooling process, leading to harvesting the heat energy to use in other equipment.

In other words, the efficiency of electronic devices depends upon their cooling system because when the temperature is high, working condition is intolerable for the whole system. Repeating unbearable conditions (working in high temperatures) leads to failure and damage to electronic units like the graphic process unit (GPU) and the central process unit (CPU). As a result, this is significant to be worked on this challenge by scholars to find an acceptable and manufacturable design for the cooling system. According to the size of electronic devices and

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Nomenclature			
x	the horizontal axis, m	φ	volume fraction of nanoparticle
y	the vertical axis, m	ε	porosity
z	the depth, m	$\dot{\gamma}$	strain rate, 1/s
$\vec{u}$	velocity vector, m/s	μ	viscosity, N/m ²
p	pressure, Pa	λ_1	the first factor of the momentum equation
T	temperature, K	λ_2	the second factor of the momentum equation
S	rate of entropy generation, W/K	<i>Super and subscripts</i>	
h	convective heat transfer coefficient, W/m ² K	out	outlet
Q	volumetric flow rate, m ³ /h	in	inlet
k	thermal conductivity, W/m.K	MF	metal foam
C_p	specific heat, J/ kg.K	s	solid
C_F	inertia factor	eff	fluid
K	permeability, m ²	wf	water fluid
A	Area, m ²	fr	frictional
V	Volume, m ³	th	thermal
<i>Greek symbols</i>		tot	total
ρ	density, kg /m ³	ave	average

the benefits of micro-scale flows, micro heatsinks are potential candidates. On the other side, this removed heat can be harvested and stored for other purposes like what is being done with phase change materials (PCMs) [5], so this is a method with two purposes; cooling the electronic chip, and heat exchange with another fluid. It is noteworthy that using microfluidics in heat transfer problems improves the heat exchange rate. As a result, because of the advantages of micro-scale flows, the applications of microchannel heat sinks are not just restricted to the cooling industry, but also energy storage (harvesting energy) like solar collectors.

However, they are still associated with problems in thermal performance and need to be developed and optimized. So, many methods have been applied to improve the thermal efficiency of micro ducts. In this part, some recent investigations are categorized. Nanofluids were employed by many researchers as coolants in micro ducts. These superior working liquids can be made using dispersing nano-sized particles in convectonal liquids such as water and ethylene glycol (EG) [6,7]. Since the nanoparticles improved thermophysical properties of base fluids, especially thermal conductivities, lots of investigations have been done to clear the impacts of nanofluids on different applications [8–10]. Likewise, three types of nanofluids were studied in a microchannel heatsink by Al-Baghdadi et al. [11] to compare their impacts on forced convection parameters. They presented that just SiO₂/water nanofluid improved forced convective better than water, and the other nanofluids (CuO/water and Al₂O₃/water) were not superior in that geometry. Nevertheless, water was still better due to the lower pumping cost. In an oriented pin-fin heatsink, Nawaz et al. [12] empirically compared the amount of dispersed graphene oxide (GO) particles in water with EG-water as a binary fluid. It showed that nanofluids performed better than the binary fluid. Also, the best performance of the heat sink occurred at the highest rate of GOs, leading to a 19.68 % increase in Nusselt (Nu) number. Also, Gonazlez et al. [13] numerically studied the impacts of the rheological behavior of coolant on pressure loss and heat exchange in a micro pin-fin heatsink. To be clear, they modeled nine nanofluids involving Newtonian and non-Newtonian viscosity behaviors in the laminar range of Reynolds (Re) number. They concluded the factor of shear-thinning behavior of nanofluid led to improving forced convection and decrementing pressure drop. According to the pros of nanotechnology, nanofluids have attracted much attention in heat transfer problems, leading to the establishment of the nanoscale heat transfer field in thermal science. However, research in the cooling industry has not been limited to just thermal fluids, and advances in

material and manufacturing sciences contributed to this field. As a result, a wide-reaching group of studies can be seen in designing, optimizing and changing geometry either in the presence of nanofluids or in their absence. Some investigations added ribs or pin-fin and the other extended surfaces to heatsinks alongside working on the shape, size, or other geometrical parameters of extended surfaces to find a geometry with higher thermal performance. For instance, Lan et al. [14] simulated a pin-fin microchannel under a constant heat flux and studied the impacts of the high and width of pin fins on forced convection and pumping cost. They stated that the highest fin, the more Nu number, and the resistant flow. In the same way, Zhu et al. [15] modeled various configurations of ribs (rectangular, triangular, diamond, and elliptic) on forced convection of deionized water within a rectangular grooved microchannel. The outcomes represented finding the best geometry was challenging because each configuration behaved differently at various Re numbers. A numerical-based investigation was conducted by Tu et al. [16] to examine the effects of shape, number, height, and transverse pitch diameter of vertical and ring micro-ribs along with changing the rate of TiO₂ nano-additives in water (from 0 to 0.5 %) on a heat exchanger system. Their results indicated that the best volume fraction of TiO₂ particles was 0.4 % due to increasing Nu number by almost 12.93 %, while the height of the ribs could improve the heat convection by around 57.25 %. Hasan [17] modeled a rectangular channel under a constant temperature as a heatsink, then added various shapes of pin-fin to the channel to compare Dimond/water and Al₂O₃/water nanofluids with water. The results illustrated Dimond/water nanofluid was better than the other coolants and circular pin-fins performed more effectively from the thermal viewpoint. In a similar investigation, diverse configurations of pin-fins in a heat sink with mono and hybrid nanofluids were examined by Ambreen et al. [18] using a multiphase Computational Fluid Dynamics (CFD) method. They concluded that rhombus fins were more reasonable due to lower pressure drop alongside an acceptable heat transfer rate. Also, MWCNT-GNP/water nanofluid cooled the heat sink better than MWCNT/water one because of the higher thermal conductivity of GNPs. In another study, Ambreen et al. [19] numerically and experimentally investigated the influence of MXenes additives on the water in a heatsink with circular pin fins as an initiative to introduce a new potential coolant. In that work, heat transfer improved up to 40.5 % in the highest percentage of additives, but the performance evaluation criteria (PEC) of nanofluid, which has the lower percentage was greater because of the more affordable pressure loss. A heatsink including a series of zigzag microchannels with MWCNT-SiO₂/EG-water was

studied by Alnaqi et al. [20] to examine the effects of zigzag characteristics and the rate of nano-additives on heat transfer and irreversibility. They obtained that pressure drop, total entropy generation and forced convection coefficient were enhanced by increasing the percentage of nano-additives and zigzag height. Shahsavari et al. [21] scrutinized the effects of connecting holes inside a microchannel heatsink with Ag/water nanofluid as the coolant to study the first and second laws of thermodynamics. According to outcomes, the merit of applying nanofluid depended upon the Re number, while adding connecting holes was valuable to improve the thermal performance of the heatsink despite increasing all sorts of entropy generation. Besides, Aguirre et al. [22] comprehensively simulated a microchannel equipped with circular and hexagonal pin fins, vortex generators at vertical walls and various viscosity behavior of coolants to study the cooling process. They concluded that changing circular pin fins to hexagonal ones was not effective in heat exchange, but adding vortex generators improved it by around 20 % in the presence of Newtonian fluids. Also, the best case was recognized in the presence of non-Newtonian fluids, reducing pumping cost by approximately 20 % due to lower apparent viscosity. Moreover, some investigations were aimed at studying porous media (metal foams (MFs) or other types of porous materials) in micro cooling devices due to resulting in changing patterning of heat transfer and fluid flow. For instance, Shen et al. [23] numerically examined the effect of diverse configurations of porous copper foams on the forced convection of water in a microchannel. They discussed different cases to find a rational model from the thermohydraulic viewpoint. To be clear, outcomes indicated that despite a jump in pressure drop, a combined model including a Y-shape porous fin, a solid fin, and a porous layer contributed to the lowest and highest amount of thermal resistance and Nu number, respectively. Likewise, Li et al. [24] had a look at comprising silicon solid and porous ribs at various locations in a water-cooled microchannel. So, they modeled microchannels in six states in addition to the base case (empty channel). Obtained outcomes revealed adding porous silicon media led to decrementing pressure drop between 12 % and 67 % than porous ribs in various cases, creating a sharp soar in PEC. Wang et al. [25] simulated a triangular and circular cross-section tube equipped with a penetrable twisted tape as a porous medium and SiO_2 /water nanofluids to analyze the rate of heat exchange. They change the rate of nano-additives, and the shape of the holes on the twisted tape to find the best PEC for this geometry. According to their outcomes, increasing the rate of SiO_2 had a positive effect on PEC, but the best case was the triangular tube with round holes from the PEC viewpoint although it incremented the pressure loss remarkably. In an empirical study, Neyestani et al. [26] compared forced convection of Al_2O_3 /water nanofluid at two volume concentrations in a copper heatsink that was equipped with solid ribs (inline and staggered ribs) and a heatsink filled with copper balls as porous media. Results confirmed that Nu number of pure water incremented 35 % by using porous media than of which in the inline case, improved up to 43 % by applying a 0.2 % nanofluid. Delisle et al. [27], in an experimental and numerical examination, also studied a microchannel heatsink and analyzed the effects of the number of channels, filling channels with the aluminum foams, pore per inch (PPI) of the MF, the coolant (water or nanofluid). They confirm that the increasing number of channels and PPI, and using nanofluid incredibly enhanced forced convection, leading to maximizing Nu number to 135.5. There are studies that used diverse coolants (Newtonian and non-Newtonian fluids or nanofluids with different nanoparticles), geometrical changes, and shapes and fabrications of porous material, and other techniques on forced convection in microchannels [28–36].

It is worth mentioning that the literature review showed that most of the previous investigations aimed to study the working fluid or filling geometry with porous media or one microchannel in a heatsink. The present study aims to fill some gaps, and the reasons for conducting this paper are mentioned in the following. Firstly, when a bunch of microchannels is created by solid ribs on the surface of a heatsink, there are

entry and exit areas before and after the microchannel section. So, depending on the flow pattern, the velocity at each microchannel situated in the heatsink is different; if there were big entrance and exit areas, velocity was the same in each microchannel, and the symmetry condition is right (most studied models just one microchannel and extended their results for the whole heat sink). However, according to the size of the chips, having a big cooling system is impossible. The current CFD-based investigation models the entire heatsink (with entry and exit sections) with a U-shaped flow pattern (entrance and exit are both on the same side) so that the variety of the velocity pattern is considered. Secondly, since using solid materials on the surface of the heatsink or filling the whole heat sink with MFs, making an upper request for pumping power is not rational, MFs are partially used in two cases. In the first one, a microchannel section is designed with MFs (use porous fins that are attached to the heatsink surface in the flow direction), then porous metal strips are applied to make barriers against the fluid flow in the second case. After that, the results are compared with an empty heatsink to demonstrate the value of using porous blocks. Also, this CFD-based study takes advantage of a hybrid nanofluid as a high-class coolant and reports the results of entropy generation alongside forced convection for every single parameter to assess the value of applying nanofluids. Ultimately, this paper investigates the effects of using partially porous foams to introduce a new structure for a heatsink with an upper-class design from the thermohydraulic viewpoint, using porous strips. As a result, the impacts of volumetric flow rate, the permeability of porous media, and nanoparticles volume concentration at various designed structures are examined to find the best case according to PEC and look at entropy generation results.

2. Problem description and governing equations

This study aims to simulate proposed designs using MFs for a heatsink to find a geometry with higher efficiency. The schematics of the base case and proposed structures are in Fig. 1. The fluids, including water and CuO-Mg/water nanofluid in volume fractions of 1 % and 3 %, are chosen as coolants. The thermophysical properties of coolants and solid and porous materials can be found in Table 1 [37]. All geometries are cooled by a U-shaper pattern flow. A $20Kw/m^2$ heat flux is applied to the bottom surface of the heatsink by the heat source, and the coolant enters at $T = 300$ K to remove the heat flux distributed by conduction heat transfer to the whole aluminum body. To study the effects of inlet velocity on parameters, fluid flows in three volumetric flow rates ($\dot{Q}_1 < \dot{Q}_2 < \dot{Q}_3$) are examined (see Table 2). Also, The porous regions in case (B) and case (C) are studied with various aluminum MFs whose characteristics are presented in Table 3. Fig. 2 shows the drawings to demonstrate the dimensions.

To model the computational domain, the fluid flow in the entire domain (transparent and MFs) is assumed in the laminar regime according to the volumetric flow rate. The simulation is conducted with fluid flow in steady-state, incompressible, constant thermophysical properties, three dimensions, and single-phase conditions. Also, the porous zones are isotropic, homogenous, and in thermal equilibrium with fluid. The governing equations based on the mentioned assumptions are as below:

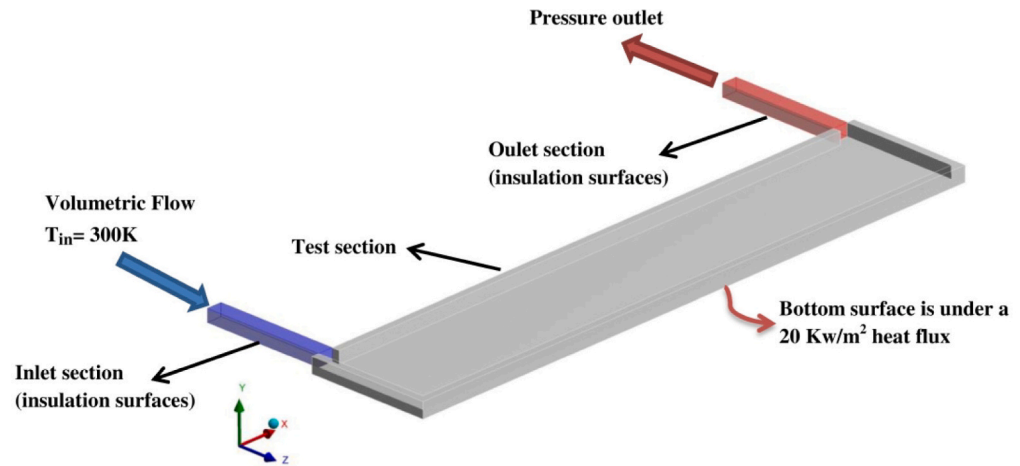
Continuity equation [38]:

$$\nabla \cdot (\rho_{wf} \vec{u}) = 0 \quad (1)$$

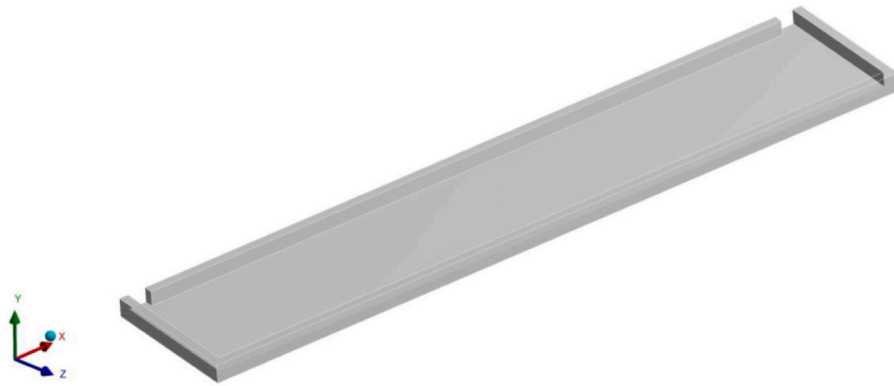
where ρ and $\vec{u}$ are density and velocity vector. The subscript wf is working fluid.

Momentum equation [38]:

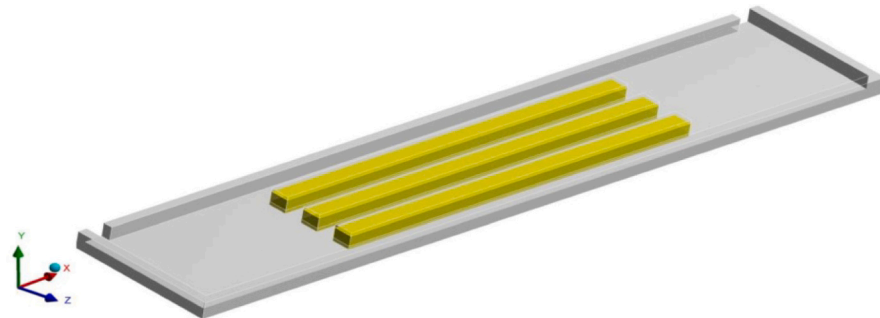
$$\frac{\rho_{wf}}{e^2} (\vec{u} \cdot \nabla) \vec{u} = -\nabla p + \mu_{wf} \nabla^2 \vec{u} - \lambda_1 \frac{\mu_{wf}}{K} \vec{u} - \lambda_2 \frac{\rho_{wf} C_F}{\sqrt{K}} |\vec{u}| \vec{u} \quad (2)$$

Main
schematic

case (A)



case (B)



case (C)

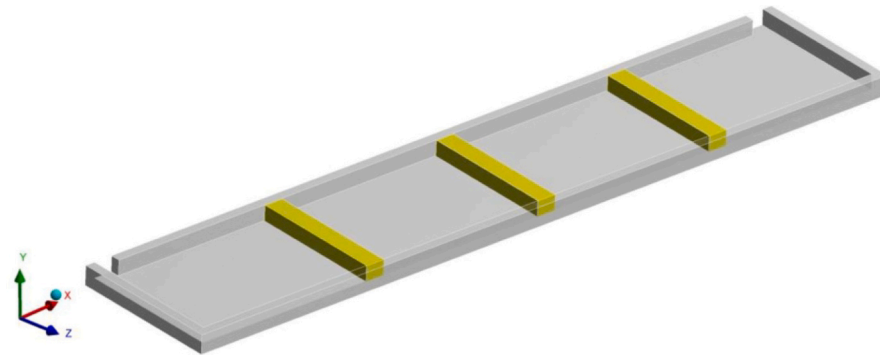


Fig. 1. The schematics of the simulated problems.

where ε , p , μ , K , and C_F are porosity (0.98 and 1 for porous and solid regions), pressure, viscosity, permeability and inertial coefficient. Also, λ_1 and λ_2 are both 0 and 1 for porous and clear zones.

The energy equation for fluid zone [38]:

$$\nabla \cdot (\rho_{wf} \vec{u} c_{p, wf} T_{wf}) = \nabla \cdot (k_{eff} \nabla T_{wf}) \quad (3)$$

where C_p , T and k are specific heat capacity, temperature and thermal

Table 1

The reported thermophysical properties of working fluids [37].

Fluid	ρ (kg/m ³)	C_p (J/kg.K)	k (W/m.K)	μ (Ns/m ²)
water	997.9	4182	0.6094	0.001003
1 % nanofluid	1008.25	3980.25	0.6721	0.001103
3%nanofluid	1151.23	3612.25	0.7125	0.001385

Table 2

The number of volumetric flow rates in this study.

Q_1 (m ³ /h)	Q_2 (m ³ /h)	Q_3 (m ³ /h)
0.06515×10^{-3}	0.227952×10^{-3}	0.553608×10^{-3}

Table 3Characteristics of aluminum MFs with an assumed in this study at $\varepsilon = 0.98$.

	K	C_F
MF ₁	1.0236×10^{-9}	0.1473
MF ₂	1.0236×10^{-11}	0.1473

conductivity. The eff subscription shows the effective thermal conductivity when the flow entrances to MFs, which is [31]:

$$k_{eff} = \varepsilon k_{wf} + (1 - \varepsilon)k_{mf} \quad (4)$$

where mf indicates metal foam.

The energy equation for the aluminum body [39]:

$$\nabla^2 T_s = 0 \quad (5)$$

where s is solid (the body of the heatsink).

The mean convective heat transfer coefficient is [39,40]:

$$h_{mean} = \frac{1}{A_{hot}} \int \frac{q''}{(T_{surface,mean} - T_{bulk,mean})} dA_{hot} \quad (6)$$

where

$$T_{bulk,mean} = \frac{T_{inlet} + T_{outlet}}{2}$$

where A_{hot} is the area of hot surfaces.

The PEC_1 is defined to evaluate the effects of MFs [21]:

$$PEC_1 = \frac{\frac{h_{mean,case(B)or(C)}}{h_{mean,case(A)}}}{\frac{\Delta p_{mean,case(B)or(C)}}{\Delta p_{mean,case(A)}}} \quad (7)$$

Where

$$\Delta p = p_{in} - p_{out}$$

The parameters in PEC_1 are calculated in the same ϕ and Q .

The PEC_2 is defined to evaluate the effects of the hybrid nanofluid [21]:

$$PEC_2 = \frac{\frac{h_{mean,\phi=1\%or3\%}}{h_{mean,\phi=0}}}{\frac{\Delta p_{mean,\phi=1\%or3\%}}{\Delta p_{mean,\phi=0}}} \quad (8)$$

The parameters in PEC_2 are calculated in the same case and Q .

To analyze the entropy generation, the following equations are used:

The volumetric thermal entropy generation rate is [21,34,41]:

$$S_{th} = \underbrace{\int \frac{k_{eff}}{T_{wf}^2} (\nabla T_{wf})^2 \cdot dV_{wf}}_{fluid} + \underbrace{\int \frac{k_s}{T_s^2} (\nabla T_s)^2 \cdot dV_s}_{Solid} \quad (9)$$

The volumetric frictional entropy generation rate is [21,31,34,41]:

$$S_{fr} = \int \frac{\mu_{wf}}{T} \dot{\gamma} + C_1 \frac{\mu_{wf}}{TK} |\vec{u}|^2 dV_{wf} \quad (10)$$

where $\dot{\gamma}$ is the shear rate and C_1 is 1 and 0 in porous and free regions.

Total volumetric entropy generation rate:

$$S_{tot} = S_{fr} + S_{th} \quad (11)$$

3. Numerical approach, grid independence, and validation

A simple algorithm based on the finite volume method (FVM) is used to connect the pressure and velocity fields [42,43]. The convection and diffusion terms are discretized by second-order upwind [42,43]. Also, to model the behavior of fluid flow in MFs, the Darcy equation extended by Forchheimer and Brinkman [44] is applied as source terms in momentum equations. Finally, the effective thermal conductivity is calculated parallelly by using Eq. (4). To show the independency of results to the generated grid, a grid study in the critical state was done, which can be seen in Table 4. The grid with 2,682,953 nodes was chosen.

To do the validation section, the experimental work of Hetsroni et al. [45] was simulated. They examined a heatsink with a stainless steel porous insert. The coolant was water, investigated in various Re numbers. Fig. 3 indicates the results of the current simulation and Hetsroni et al. [45] for diverse Re numbers in the presence of the MF with $K = 9.6 \times 10^{-10}$, $C_F = 0.99 \times 10^{-5}$ and $\varepsilon = 0.44$. The maximum error in the figure confirms that the outcomes in this simulation are reliable.

4. Results

Fig. 4 indicates the velocity contours for various cases at the same Q and ϕ . Using MFs changes the velocity distribution. In case (A), the flow enters the heat sink like a jet and bop to the opposite surfaces, then goes to the exit section, so the velocity has a higher amount near the opposite surface of the entry. The velocity is distributed better in case (B) thanks to the presence of MFs. MF₂ changes the velocity distribution widely because of more resistance against the flow (see Table 3). As can be observed, the velocity increases between porous areas (like a micro-channel section). Likewise, the velocity distribution experiences a change in case (C), where MFs perform like barriers against the flow. The flow hardly enters barriers, leading to having a flatter velocity distribution along the heatsink, amplifying by using an MF with lower permeability (MF₂).

The temperature contours for various cases at the same Q and ϕ are depicted in Fig. 5. The temperature field directly relates to the velocity. So, the temperature is lower where the flow has a higher velocity in case (A). In case (B), although the temperature increments in the middle of the porous blocks, the temperature field is uniform in the whole heatsink due to better partition of the coolant by using MFs. Applying an MF that creates more resistance against the flow increases temperature where the porous blocks are located and decreases in the transparent regions. This story also works for case (B), which tries to improve the temperature distribution by porous barriers. To be exact, they hold the coolant more in the heat sink, incrementing the time for heat exchange.

Fig. 6 shows the temperature field in the solid body of the heatsink for all cases at a special Q and ϕ . The temperature in case (A) is very high because of the low time for heat transfer between the coolant and the hot surfaces. In case (B), MF₁ performs better than MF₂, rooting in penetrating more flow in the porous blocks. To be clear, MFs improved the heat exchange for two main reasons: changing the velocity field and improving the effective thermal conductivity in porous zones for the coolant. For this case, the second one works more, so this makes sense that MF₁, which is more permeable, performs better than MF₂. However, the first reason is more influencer in case (C). As a result, the aluminum body is cooler in the presence of MF₂ due to keeping flow much more in the duct.

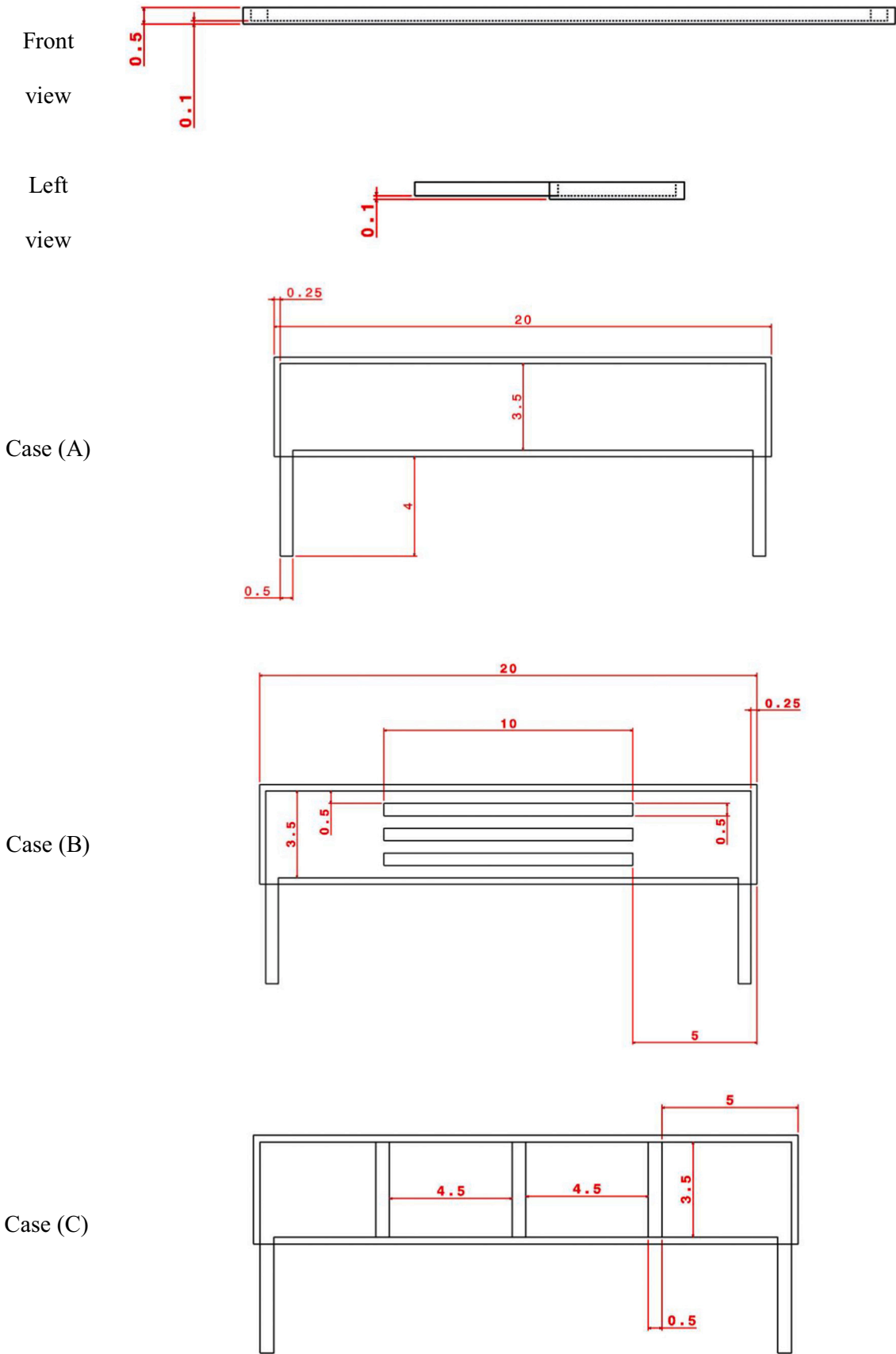


Fig. 2. Drawings for different cases with dimensions in millimeters (mm).

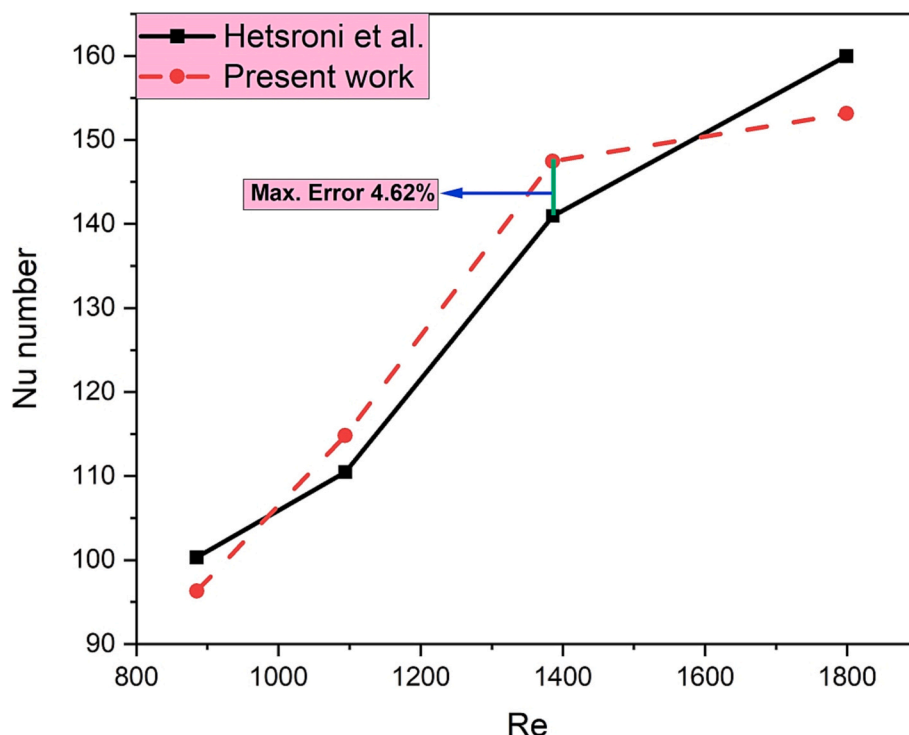
Table 4Grid study at Q_3 , $\phi = 3\%$ and MF_2 for Case 3.

Nodes	h_{ave}	Error	Δp	Error
681,774	9050.95	5.27 %	24,877.32	2.98 %
1,306,771	8597.84	2.19 %	25,641.49	1.2 %
2,682,953	8413.58	0.95 %	25,952.92	0.11 %
4,840,423	8334.03	–	25,981.11	

T_{mean} , which shows the uniformity of the temperature distribution is represented in Fig. 7-a versus all cases in the current research. Increasing the percentage of hybrid particles decrements T_{mean} , indicating a premiere temperature uniformity by using nanofluids. It happens because of the superior thermal conductivity of nano-coolants, making the temperature profile flatter. Also, MF_1 contributes to diffusing temperature in a better way than MF_2 in case (B), but it does not work for case (C) (an MF with more resistance is better (MF_2)). These results conclude because case (A) uses the benefits of MFs' thermal conductivity, and case (B) uses the pros of changing fluid flow patterns by the resistance of MFs. To explain them simply, using porous media with higher permeability cause increasing the rate of crossing flow in this zone, increasing the effective conductivity. However, when a design takes advantage of a porous medium because of the solid behavior, an MF with less permeability is preferable. Moreover, the temperature is more uniform with rising the volumetric flow rate because the more velocity, the more cooling. Increasing velocity leads to making a delay in the formation of hydrodynamic and thermal boundary layers. The maximum temperature in the aluminum part of the heatsink is shown in Fig. 7-b. T_{max} indicates how the design affects the cooling of the heatsink's body. In all states, T_{max} is reduced by increasing ϕ due to the advantage of nanofluids (higher thermal conductivity leads to enhancing heat exchange). While MF_1 is added to the heatsink in case (B), T_{max} experienced a more decrease than MF_2 . However, for case (C), the outcomes seem different, showing MF_2 performs better in decreasing T_{max} a little bit. Also, Incrementing Q contributes to a sharp fall in T_{max} . For instance, T_{max} falls from the rage of 323–325 to the rage of 306–307.5. (see the explanation in Fig. 7-a to find more detail about the reasons).

Fig. 8-a illustrates the amount of h_{mean} for all states in this study versus variables. Increasing Q develops the convective heat transfer because of changing velocity field, affecting both boundary layers along the heatsink, so a boom can be seen with incrementing velocity in h_{mean} . It is true that flow with high velocity cools the heatsink better. Likewise, using an upper-class coolant like nano-coolants amplifies these impacts due to more heat exchange, which is clear in all cases and volumetric flow rates. However, although adding MFs improves the h_{mean} , the number depends on the permeability and arrangement. Case (B) with MF_1 improves h_{mean} more preferable than MF_1 because the flow has an opportunity to penetrate better in porous metal mediums and contributes to enhancing the effective thermal conductivity. Nevertheless, there is no significant difference between the results of MF_1 and MF_2 though the results experience a small jump for MF_2 in case (C). In this case, an MF having lower permeability performs better because it navigates the flow to the heatsink's surfaces, boosting velocity in those areas and h_{mean} . It is worth mentioning that both cases profit from being changed velocity field and improving thermal conductivity; nonetheless, the first benefit is more witnessed in case (C) and the second one is in case (B). Case (B) in the presence of MF_1 fulfills increasing h_{mean} better though a drop at Q_2 . The mentioned case can increase h_{mean} approximately from 4000 to 8000 depending on ϕ and Q . The values of pumping costs for diverse research states versus Q are provided in Fig. 8-b. Incrementing velocity and making more turbulence in ducts cause demanding more pumping power, which happens with incrementing Q . The viscosity of nanofluid develops alongside rising the rate of nano-additives, making more pumping cost. Inserting porous materials into the bottom surface of the heatsink changes the velocity pattern, leading to incrementing pressure loss, which elevates using a lower permeable MF. The pumping cost sharply soars for case (C) with MF_2 . The porous blocks show behavior like solid materials, and entering coolant is strict.

Fig. 9-a discloses the number of PEC_1 that compares the thermal performance of inserting MFs in the heatsink (case (B) and (C)) to empty case (case (A)) regarding the pumping cost at the same ϕ and Q . The results that are more than one show an improvement. The outcomes reveal that using MF_1 is better than MF_2 because it needs less pumping

**Fig. 3.** The comparison of Nu number between this simulation and Hestroni et al. [45].

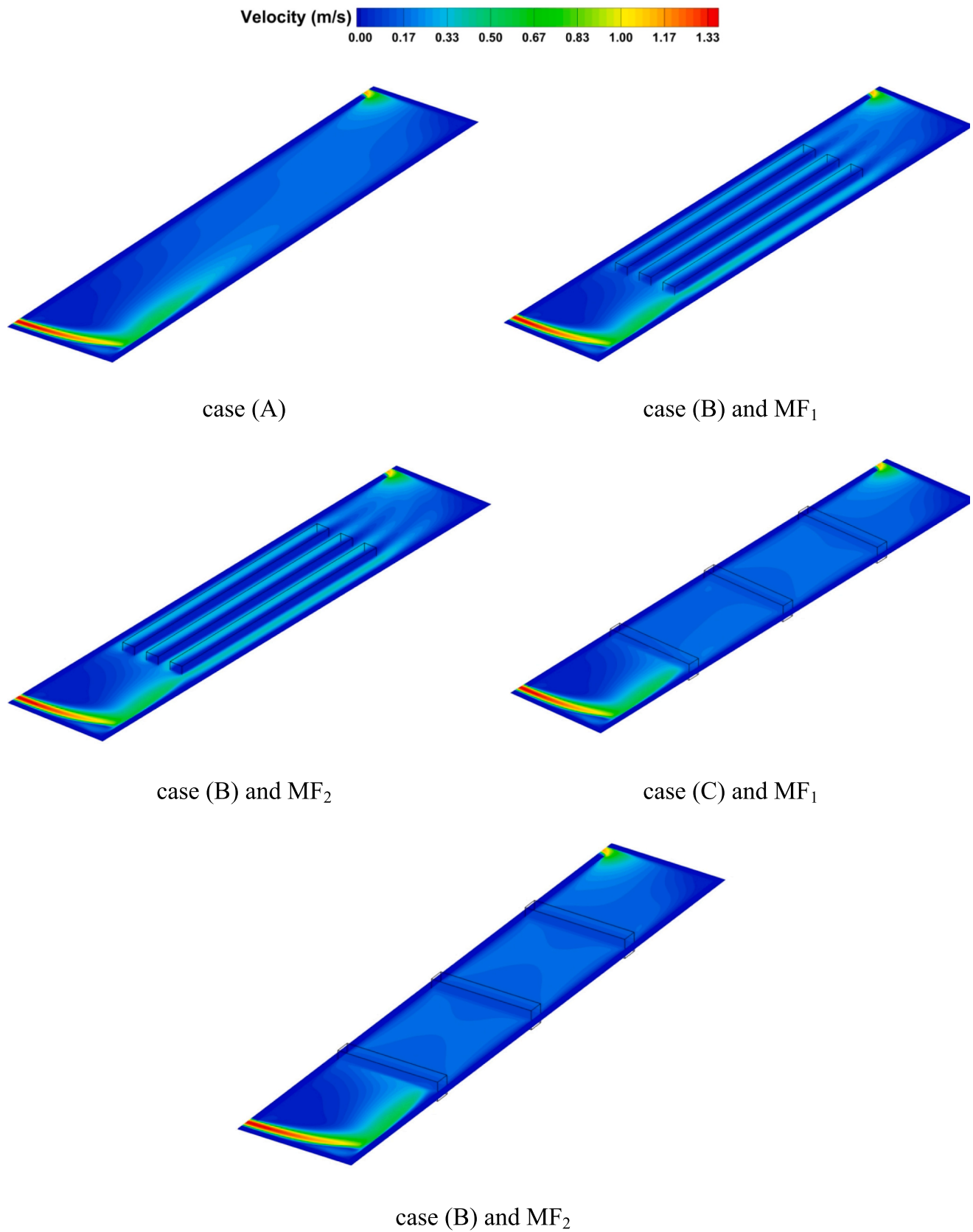


Fig. 4. The velocity contours on plane $z = 0.3$ mm for various cases at Q_3 and $\varphi = 3\%$.

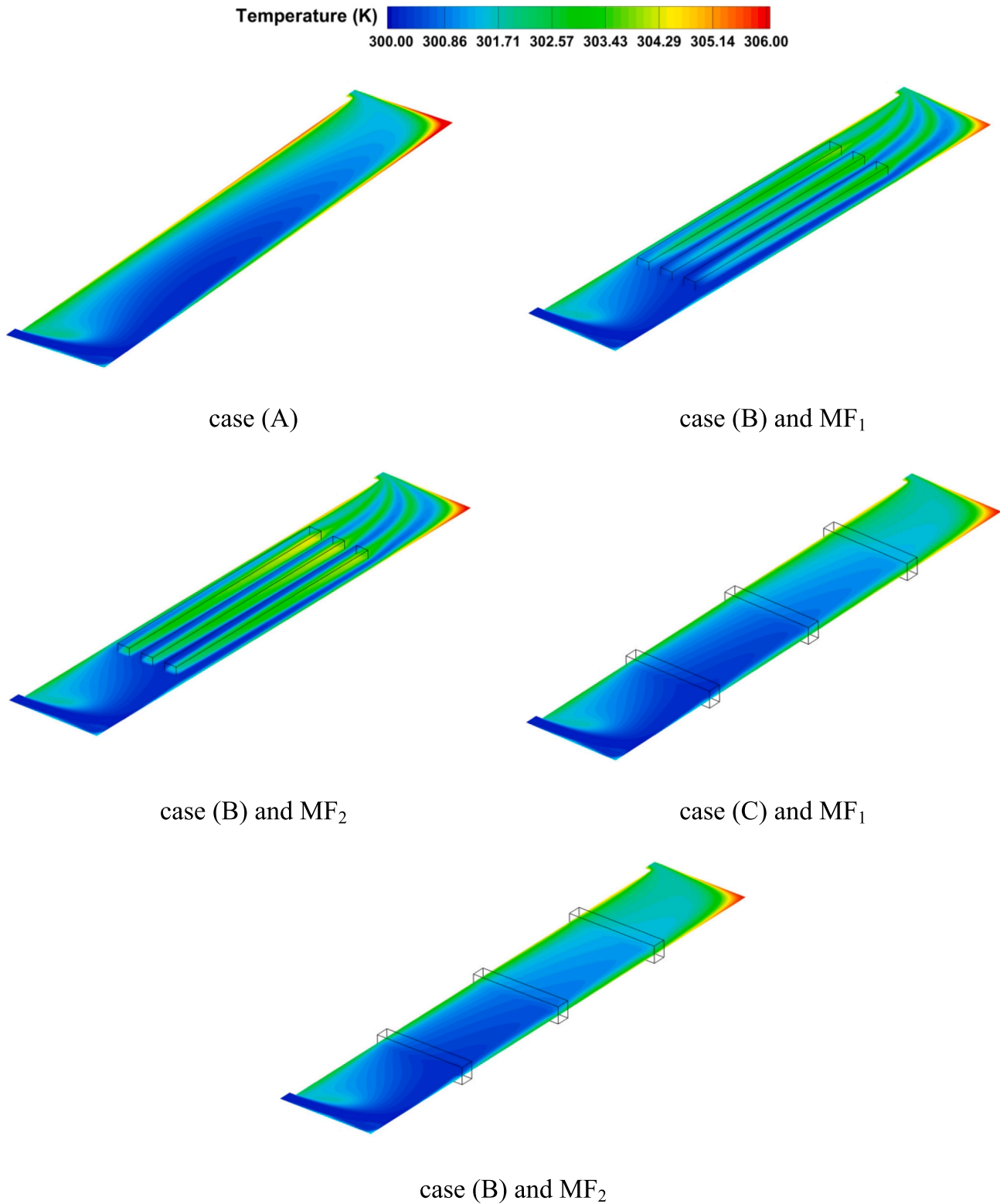


Fig. 5. The temperature contours of the fluid zone on plane $z = 0.3$ mm for various cases at Q_3 and $\phi = 3\%$.

power. On the other hand, although MF₂ provides acceptable performance in case (B), it is not affordable in case (C) due to irrational pressure drop (PEC is around between 0.6 and 0.7). Also, nanofluids decrease heat sink performance in the presence of MFs, so water should be applied as the coolant in cases (B) and (C). All in all, there is still a tight competition between cases (B) and (C) in the presence of MF₁, which hinge on Q . To introduce the higher number of PEC₁, case (B) at

Q_3 in the presence and absence of MF₁ and nanofluids overrides the other cases and indicates a number more than 1.5.

At the same Q , case and MF, the performance of adding nano-additive to the cooling process in the heat sink is considered, and outcomes are shown in Fig. 9-b. The numbers indicate that adding nanoparticles is not advised despite nanofluid helping the cooling process, enhancement of viscosity increments the friction between the fluid and

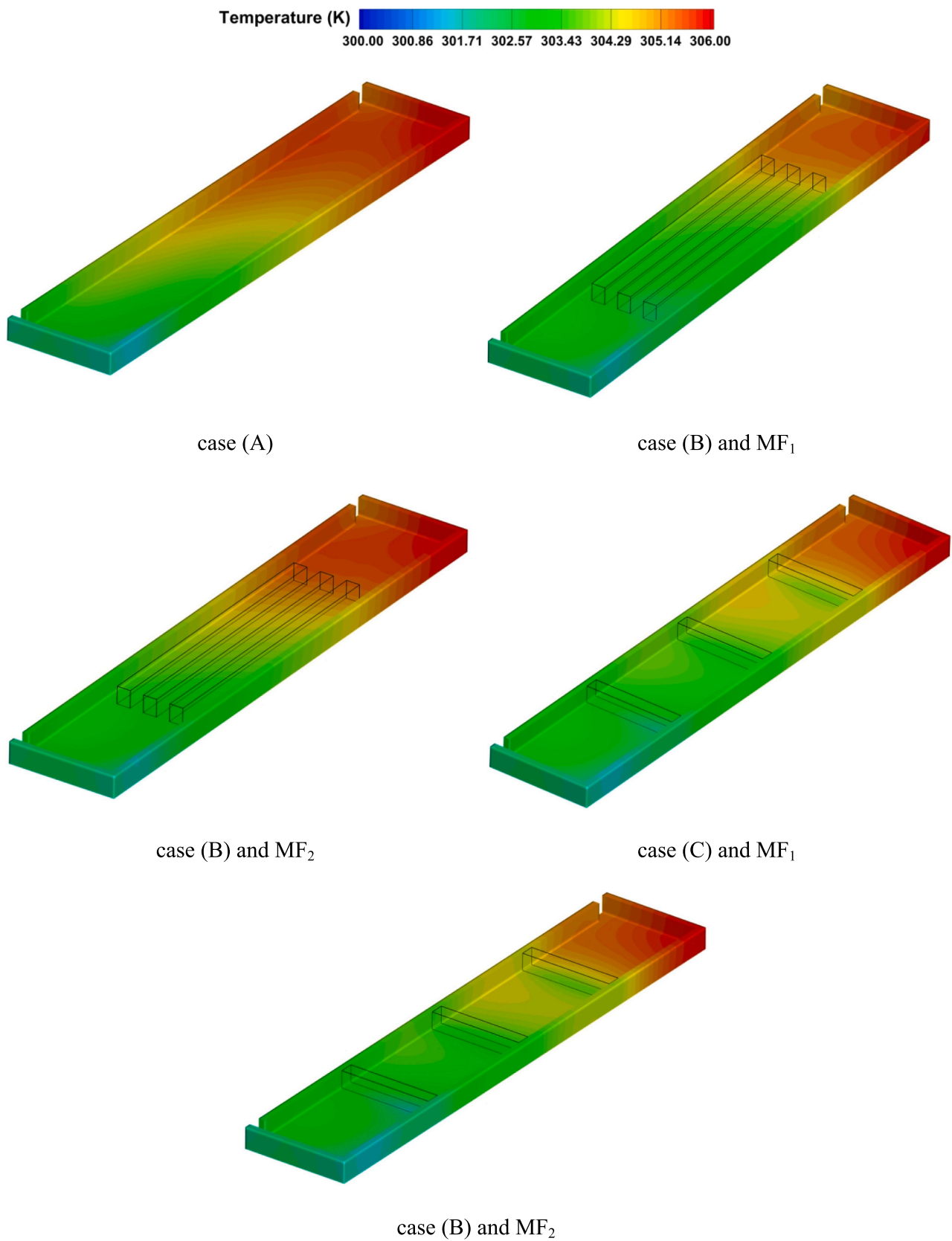
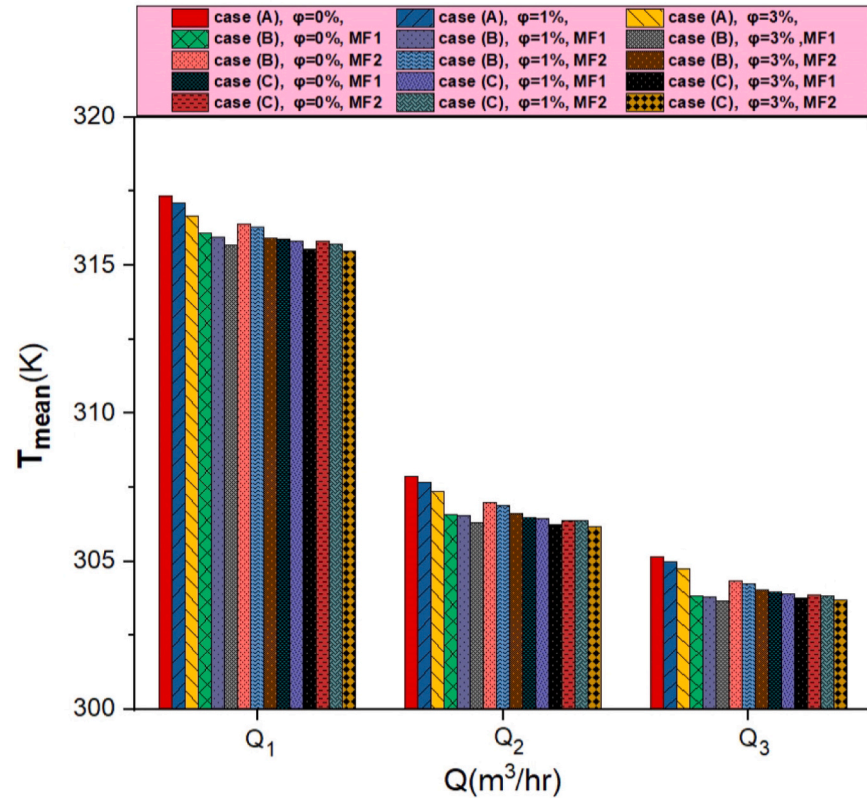


Fig. 6. The temperature contours of the solid zone for various cases at Q_3 and $\phi = 3\%$.

a)



b)

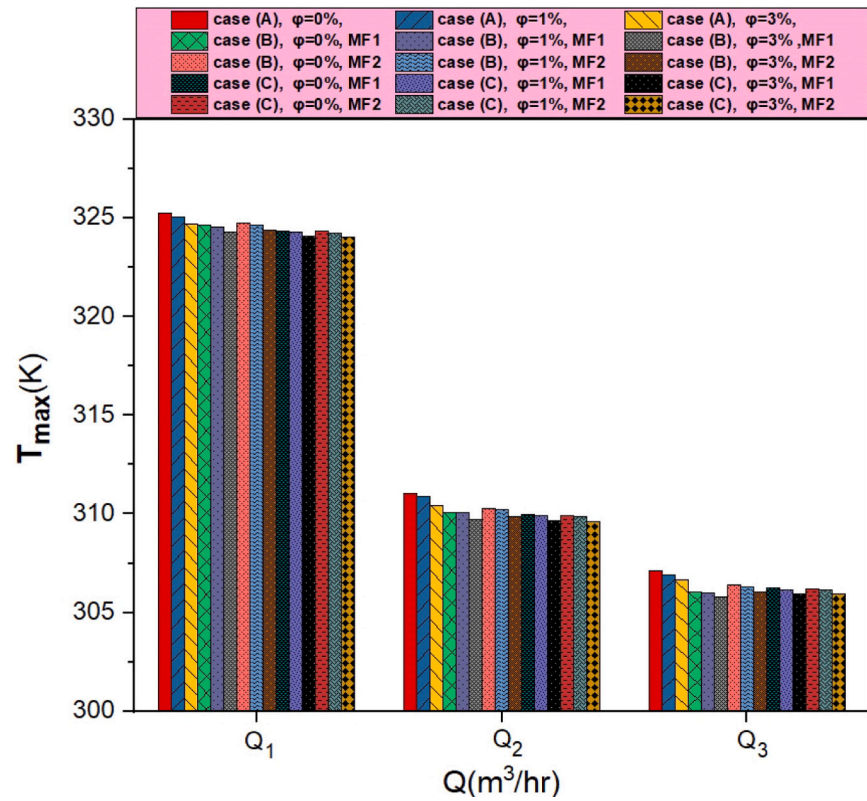
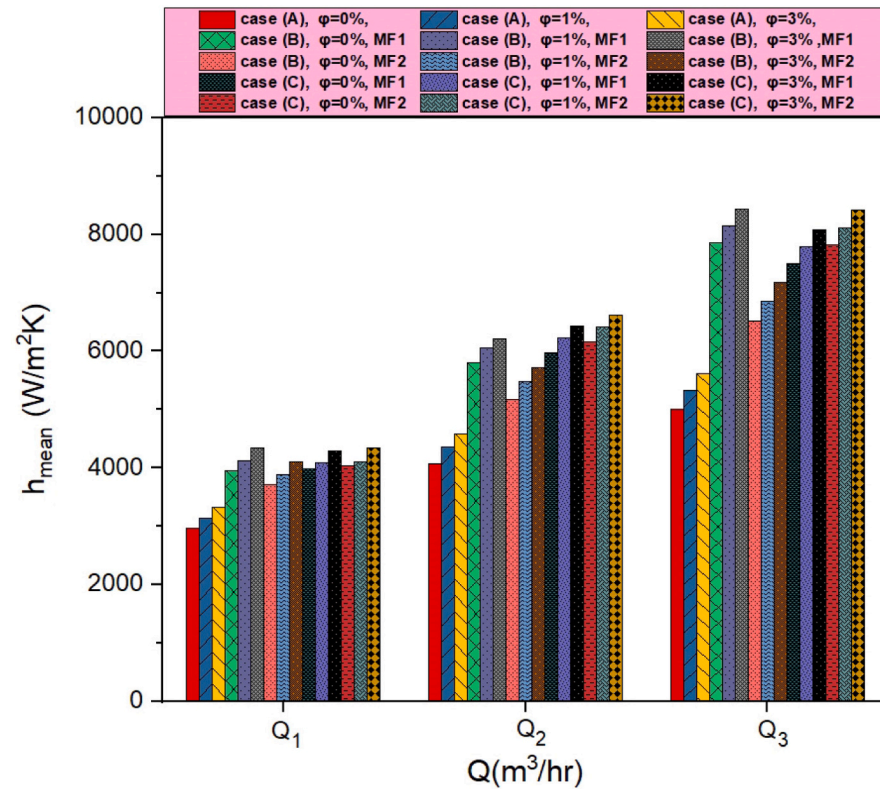


Fig. 7. The a) mean b) maximum temperature in the solid part for all states.

a)



b)

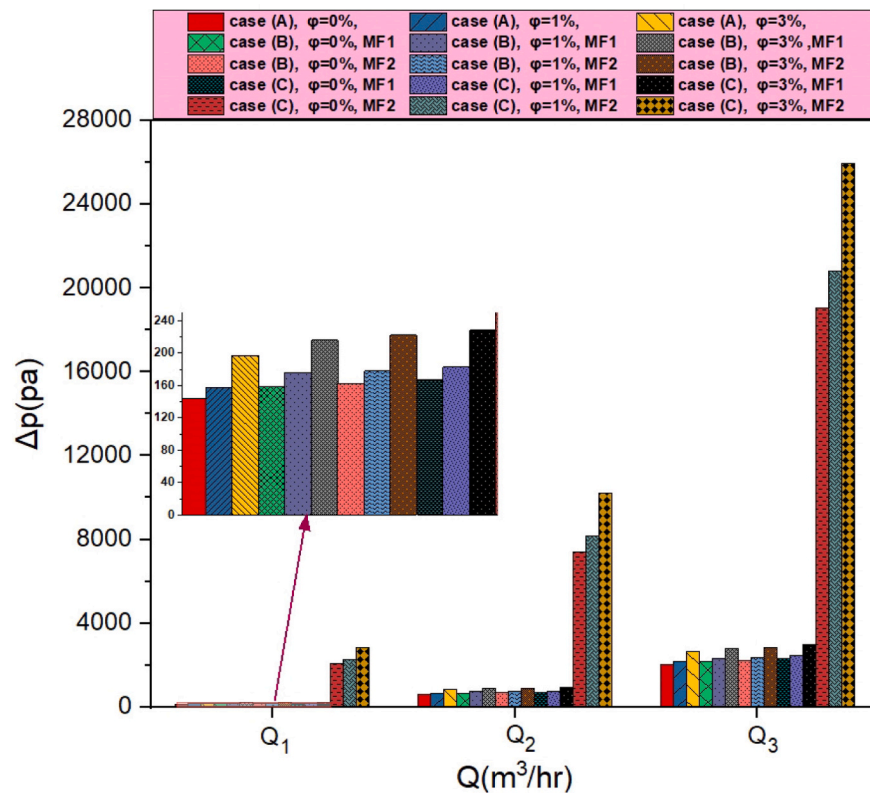


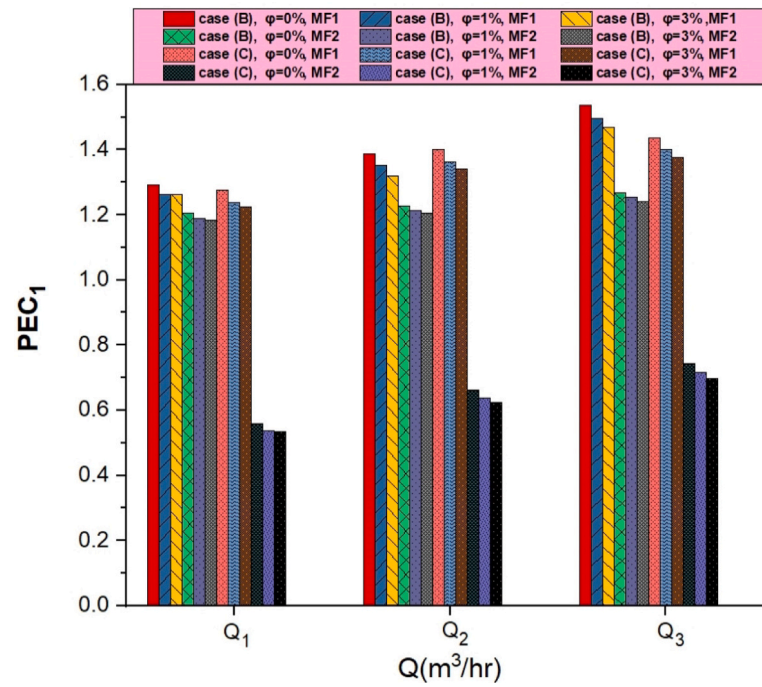
Fig. 8. The variations of a) mean convective heat transfer coefficient b) pressure drop for all states.

solid parts. Likewise, developing their percentages in the base fluid makes the performance worse. 1 % hybrid nanofluid performs reasonably just in the absence of MFs and the highest number of Q . Using hybrid nanofluid can decrement the heatsink's performance by

approximately 0.22 due to demanding more pumping power than water.

The outcomes of thermal entropy generation rate (ThEGR) for the fluid and solid parts are indicated in Fig. 10-a. The better temperature uniformity in both zones decreases ThEGR by increasing Q and ϕ . Also,

a)



b)

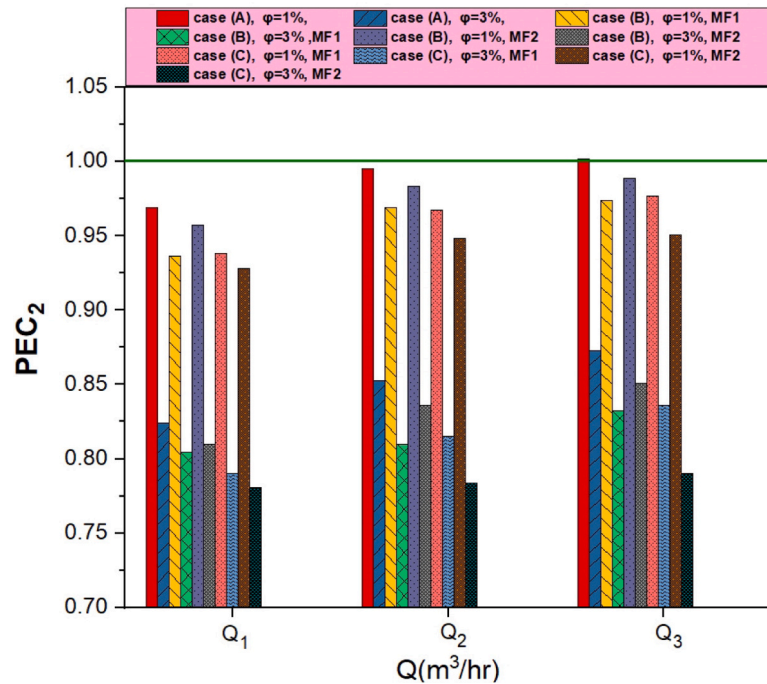


Fig. 9. The variations of a) PEC_1 b) PEC_2 .

adding porous blocks in all cases reduces ThEGR more remarkably than in case (A) because porous mediums keep the temperature steady in the heatsink. Case (B), MF₁ and case (C), MF₂ decrement ThEGR more than of that in the same case with the other MF.

Fig. 10-b illustrates the frictional entropy generation rate (FEGR) for all states in this simulation. FEGR increases by adding nano-additives and increasing volumetric flow rate in all cases due to more friction between the coolant and the aluminum body, rooting in changing viscosity and velocity gradient, respectively. An enhancement can be seen in FEGR by inserting MFs for two key reasons: The first one is that changing velocity gradient and profile in the ducts and the second one is that a term is added to FEGR equation by using MFs. Also, equipping the

heatsink with a lower permeable MF increases the first term of the FEGR equation, but it reduces the second term due to less penetration. So, the results do not have a special pattern. The outcomes of total entropy generation rate (TEGR) are given in Fig. 10-c versus diverse cases. The results indicate that ThEGR dominates the TEGR because it has a similar pattern in this figure.

5. Conclusion

This study, due to the applications and benefits of micro-scale flows for harvesting energy and increasing heat exchange, was looking to introduce a new design in a heat sink using MFs partially in order to find

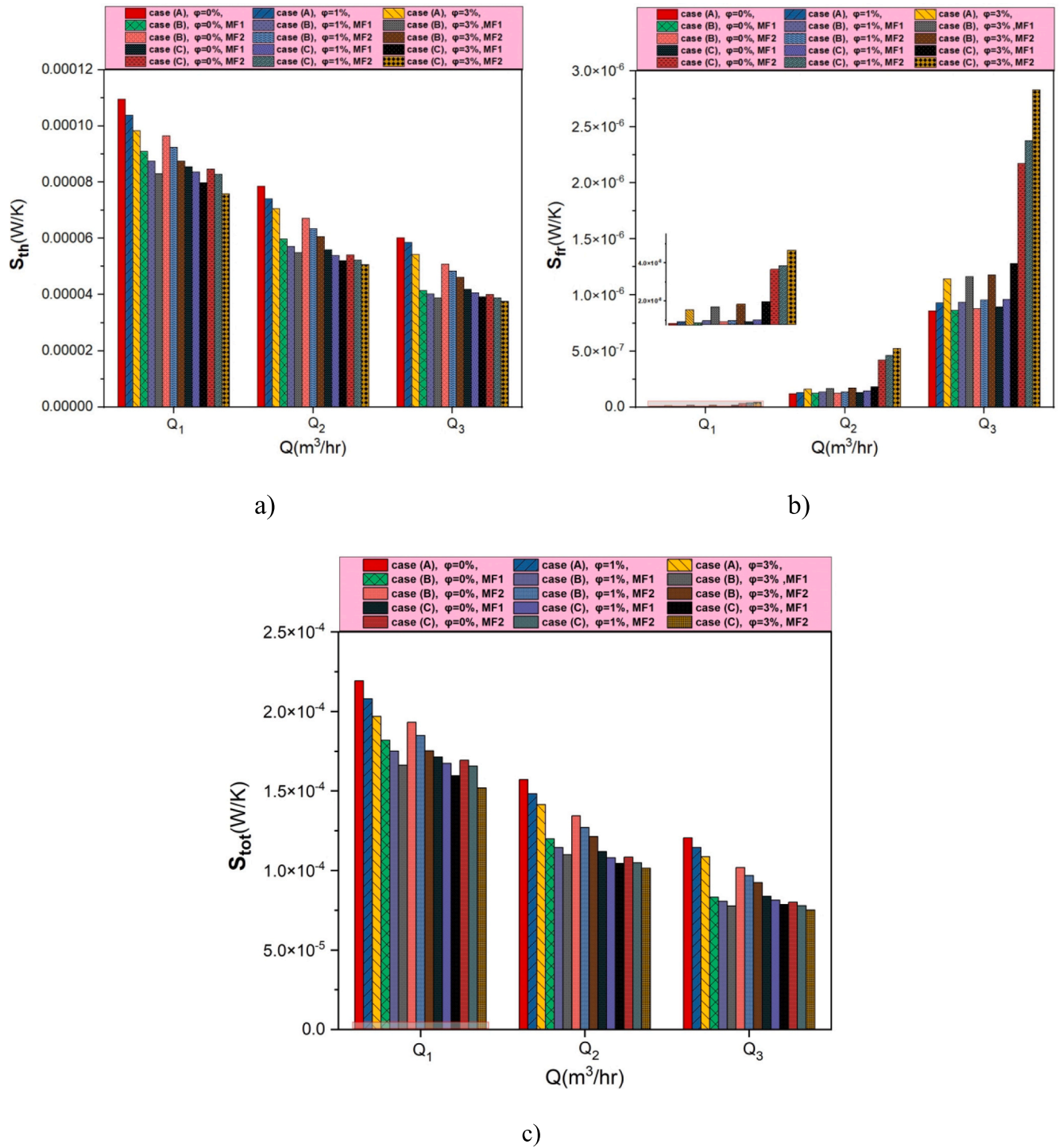


Fig. 10. a) Thermal b) frictional c) total entropy generation rate for all states.

a compromise between increasing heat transfer and pressure drop. To find this optimal geometry and condition, a numerical investigation was conducted to show the effects of applying MFs and hybrid nanofluids on the cooling process in a heatsink. The working fluid cooled the heatsink under a constant heat flux by forced convection. On the other side, the conduction heat transfer happened in the aluminum body of the heat-sink. Hybrid nanoparticles, including MgO and CuO, were added to the base coolant, water, to synthesize a superior working fluid. After that, two geometries entitled cases (B) and (C) were equipped with MFs in different arrangements. Case (B) had three porous blocks in the middle section (in the direction of the flow) to build a microchannel section that

was made by MFs. Porous blocks were located like barriers along the heatsink in case (C). In each arrangement, the influence of the permeability of MFs was studied. All cases were investigated in various volumetric flow rates, 0, 1, and 3 % of hybrid nano-additives, the permeability of MFs, then thermohydraulic variables and entropy generation were reported.

The main findings in this simulation are:

- Increasing the volumetric flow rate and the percentages of nano-additives contributed to improving the forced convection and an increment in pressure drop.

- Adding MFs to the heatsink improved the convective heat transfer coefficient than the empty heatsink so that the first arrangement (case (B)) and the second arrangement (case (C)) improved it from more than 20 % to just under 60 % and from more than 25 % to just under 60 %, respectively, which depended on the MF and Q.
- In the presence of porous media, pressure loss to pump the coolant increased between 8 and 12 % for case (B). Also, reducing the permeability of porous media affected the pressure loss in case (C) substantially. As a result, in the presence of MF₁ (more permeability), pressure loss enhanced in the range of 10–15 %, but it was in the range of 900–1450 % with inserting MF₂.
- PEC₁ of the heatsink increases approximately between 1.25 and 1.55 and between 1.18 and 1.22 with inserting MF₁ and MF₂ in case (B).
- PEC₁ of the heatsink enhanced around in the range 1.22–1.38 with MF₁ and reduced in the range of 0.52–0.7 with MF₂ for case (C).
- PEC₂ shows that nanofluids were unsuitable by adding MFs and dramatically reduced the heatsink's performance.
- Despite increasing the FEGR by applying nano-additives and MFs, TEGR experienced a decrement because of the higher portion of TheGR that was decremented by those.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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